

Random Signal Theory–EEE 554

Lecture Notes 2

Discrete and Continuous Random Variables

[Reading: Kay, Sections 5.3, 5.4, 5.8, 10.3, 10.4, 10.6]

Concept of random variable (RV)

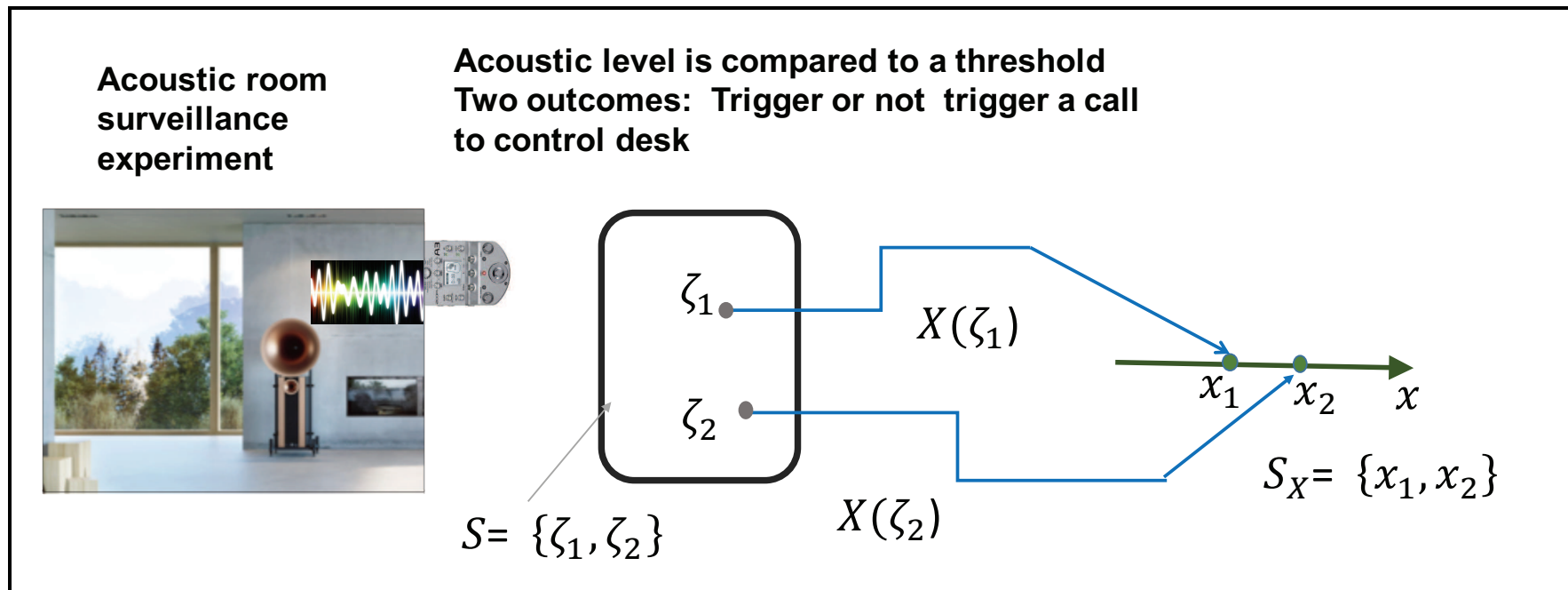
Discrete RV (DRV) [Kay, Chapter 5]

Continuous RV (CRV) [Kay, Chapter 10]

For an experiment with sample space $\mathcal{S}$ and outcomes denoted by $\zeta_i, i = 1, 2, \dots$ a random variable (RV) X is a function that maps the outcomes ζ_i to discrete or real numbers x_i (values that the function takes)

Domain of the function is the sample space $\mathcal{S} = \{\zeta_1, \zeta_2, \dots, \}$ and range of the function is $\mathcal{S}_x = \{x_1, x_2, \dots, \}$

Example: **Acoustic intruder detection**



Discrete RV (DRV):

The sample space S_X of a DRV X is a finite or countably infinite set of integers ($\mathbb{Z}$)

If $\{x_1 \in S_X : x_1 \in \mathbb{Z}\}$, then we can consider $\Pr(X = x_1)$, which is the probability that X takes the integer value x_1 , or other probabilities such as $\Pr(X \leq x_1)$ and $\Pr(X < x_1)$

Statistical measures that provide useful information for DRVs:

probability mass function (PMF), cumulative distribution function (CDF),
mean, average, variance, moments, characteristic function

Continuous RV (CRV):

The sample space S_X of a CRV X is the real line (real numbers $\mathbb{R}$)

If $\{x_0 \in S_X : x_0 \in \mathbb{R}\}$, we can only consider $\Pr(X \leq x_0) \equiv \Pr(X < x_0)$ since $\Pr(X = x_0) = 0$

Statistical measures that provide useful information for DRVs:

probability density function (PDF), cumulative distribution function (CDF),
mean, average, variance, moments, characteristic function

Mixed RVs: CRV that can take some discrete values (piecewise continuous)

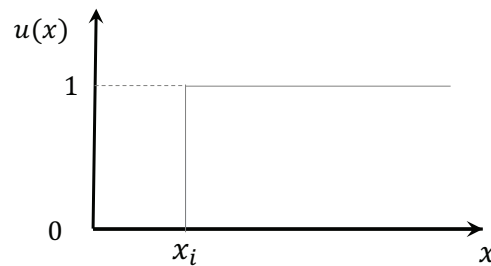
Discrete RV (DRV), Cumulative Distribution Function (CDF) [Kay, Section 5.8]

Consider a DRV X with range values in the countable set $S_X = \{x_1, x_2, \dots, x_M\}$, $x_i \in \mathbb{Z}$

► CDF of X provides the probability of ranges of values of x and is defined as

$$F_X(x) = \Pr(X \leq x) = \sum_{i=1}^M \Pr(X = x_i) u(x - x_i)$$

where the unit step (in the continuous domain) is $u(x - x_i) = \begin{cases} 1, & x > x_i \\ 0, & x < x_i \end{cases}$



► CDF is a non-decreasing, positive function and has maximum value 1

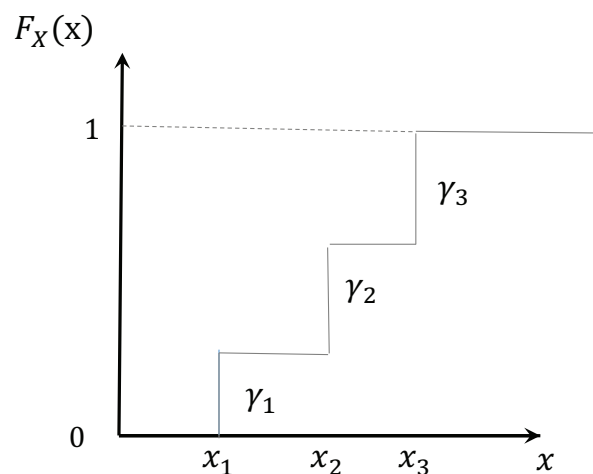
Example DRV CDF:

Let $M = 3$ and $\Pr(X = x_i) = \gamma_i, i = 1, 2, 3$

$\Rightarrow \gamma_i$ is the size of the discontinuity (jump)

The CDF of X is given by

$$F_X(x) = \sum_{i=1}^3 \Pr(X = x_i) u(x - x_i) = \begin{cases} 0, & x < x_1 \\ \gamma_1, & x_1 \leq x < x_2 \\ \gamma_1 + \gamma_2, & x_2 \leq x < x_3 \\ \gamma_1 + \gamma_2 + \gamma_3 = 1, & x \geq x_3 \end{cases}$$



Example DRV CDF:

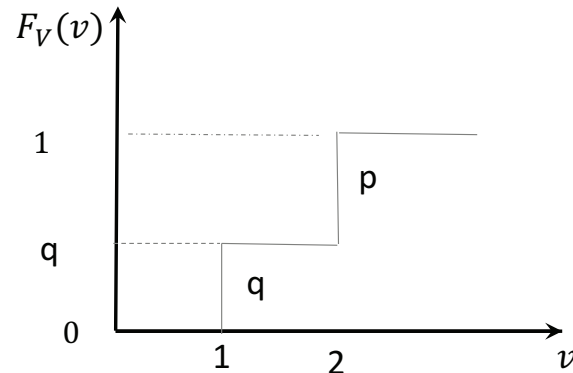
Observe the voltage V of a parity bit in a word in computer memory

assign $v = 2$ if the bit is on and assign $v = 1$ if the bit is off

assume that $\Pr(V = 2) = p$ and $\Pr(V = 1) = 1 - p = q$

For this example, the sample space is $S_V = \{1, 2\}$ and the CDF is given by

$$F_V(v) = \Pr(V \leq v) = q u(v - 1) + p u(v - 2) = \begin{cases} 0, & v < 1 \\ q & 1 \leq v < 2 \\ q + p = 1 & v \geq 2 \end{cases}$$



Using above expression for CDF, $F_V(1.5) = \Pr(V \leq 1.5) = \Pr(V = 1) = q$

$\Rightarrow$ the value x_i , that a DRV uses to map an outcome, does not have to be an integer